

Optimal Resource Allocation and Scheduling of Athletic Activities, with an Application to Triathlon

Linear and 0–1 Integer Programming models for training load, race calendar and race pacing

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Model 1 (LP) → Model 2 (0–1 MIP) → Model 3 (LP)

training load

race calendar

pacing strategy

Real data: 32 months of my own training

- Strava export: **598 activities**, 574 usable
- Banister training impulse per activity
- Fitness CTL ($\tau=42\text{d}$), fatigue ATL ($\tau=7\text{d}$)
- Today: CTL 78.1, ATL 91.5, **TSB** -13.4

r_{db}/h	easy	mod	hard
swim	43.5	94.1*	136.6*
bike	68.3	86.9	142.7*
run	81.6	102.8	166.0*

load rate per hour * theoretical (sparse data)

Model 1 — training-load allocation (LP)

Variables hours $h_{dbw} \geq 0$: 3 sports $\times$ 3 buckets $\times$ 16 weeks = 144

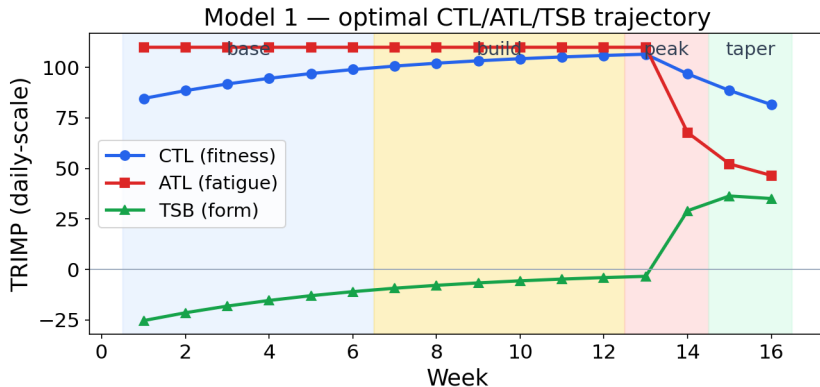
Banister dynamics, as linear equalities

$$L_w = \sum_{d,b} r_{db} h_{dbw}, \quad \text{CTL}_w = \lambda_c \text{CTL}_{w-1} + (1-\lambda_c) \frac{L_w}{7}, \quad \text{ATL}_w = \lambda_a \text{ATL}_{w-1} + (1-\lambda_a) \frac{L_w}{7}$$

Objective $\max p_T = k_1 \text{CTL}_T - k_2 \text{ATL}_T$

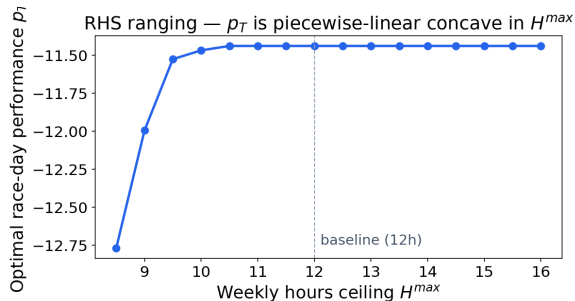
Constraints weekly hour ceilings $\cdot$ per-sport min/max $\cdot$ 80/20 polarization $\cdot$ $\text{ATL} \leq 110$ $\cdot$ ramp $\leq +10\%/ \text{week}$ $\cdot$ taper caps

Model 1 — the optimizer discovers periodization



Nobody told it to periodize. Week 14 tapers by choice.

Model 1 — duality in action



RHS ranging: the hours ceiling, 8–16 h

- piecewise-linear, concave
- slopes $1.55 \rightarrow 0.94 \rightarrow 0.12 \rightarrow 0$
- flat above **10.5 h/week**

*Recovery capacity, not free time,
is what limits this athlete.*

Model 2 — race-calendar selection (0–1 MIP)

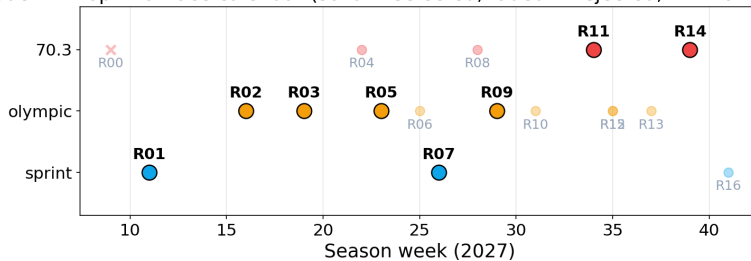
17 candidate races, binary x_i

$$\max \sum_i q_i \underbrace{(\alpha \text{ pts}_i + \beta \text{ strat}_i + \gamma \text{ enj}_i + \varphi \text{ fit}_i)}_{\text{weather-risk-adjusted value}} x_i$$

- **two knapsacks:** ≤ 2500 EUR, ≤ 12 vacation days
- **recovery spacing:** $x_i + x_j \leq 1$; ≤ 2 races/month; ≥ 1 A-race
- **readiness gate from Model 1:** projected CTL must reach the class

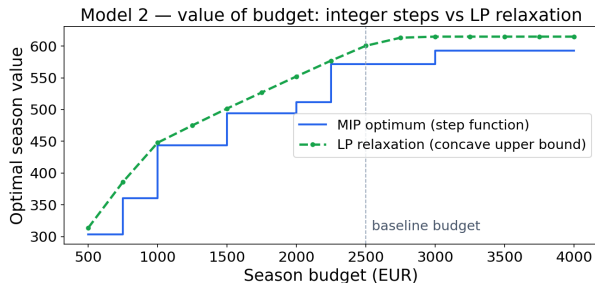
Model 2 — optimal season

Model 2 — optimal race calendar (solid = selected, faded = rejected, x = not race-ready)



8 of 17 races · value 571.7 · 2250/2500 EUR · 11/12 days

Model 2 — why Branch & Bound is needed



- LP 600.5 vs. MIP 571.7:
integrality gap 5.0%
- 8 fractional variables; the champs/70.3
conflict splits **0.5/0.5**
- **clique cuts** (Bron–Kerbosch):
600.5 $\rightarrow$ 591.0,
gap **5.04% $\rightarrow$ 3.38%**

Model 2 — the dual problem

$$\kappa_i y_{\mathcal{B}} + \delta_i y_{\mathcal{D}} + y_{\text{month}(i)} + \sum_j y_{ij} - [i \in A] y_A \geq v_i$$

a race enters only if its value covers the shadow cost of what it consumes

Dual	Meaning	Price
$y_{13,14}$	Champs vs. Costa Navarino conflict	56.40
$y_{06,07}$	Hamburg vs. Thessaloniki conflict	31.67
$y_{\mathcal{B}}$	one extra euro	0.094
$y_{\mathcal{D}}$	one extra vacation day	≈ 0

The scarce resource is calendar space — not money.

Model 2 — network-flow extension

Drop the knapsacks and the monthly cap $\Rightarrow$ only recovery spacing remains:

feasible calendars = paths in a DAG

Three methods, identical answer $z = 639.4$:

- 1 dynamic programming — longest path in topological order
- 2 **flow LP, continuous variables — integral anyway**
- 3 restricted 0–1 MIP

Totally unimodular. The knapsacks are what break the structure.

Model 3 — race pacing (LP with linearization)

Variables minutes $t_{\ell z}$: 3 legs $\times$ 5 zones = 15

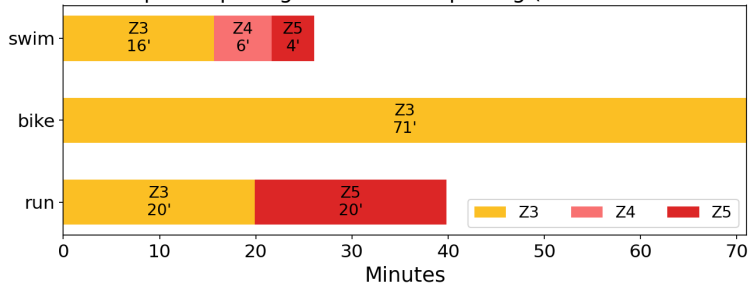
Linearization within a zone, speed and energy rate are *constants*

$$\min \sum_{\ell, z} t_{\ell z} + T_1 + T_2 \quad \text{s.t.} \quad \underbrace{\sum_z s_{\ell z} t_{\ell z} \geq d_\ell}_{\text{distances}}, \underbrace{\sum e_{\ell z} t_{\ell z} \leq k_E \cdot \text{CTL}_T}_{\text{energy budget (Model 1)}}, \underbrace{\text{Z4+ caps}}_{\text{per leg}}, \underbrace{\varphi \cdot G}_{\text{bike} \rightarrow \text{run}}$$

bike speed $\propto \phi(\gamma) \cdot (\text{power})^{1/3}$ — $\phi(\gamma)$ from the power balance, **derived, not fitted**

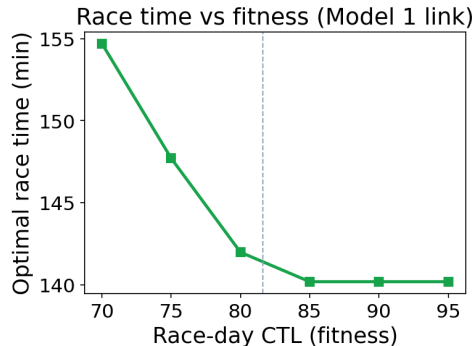
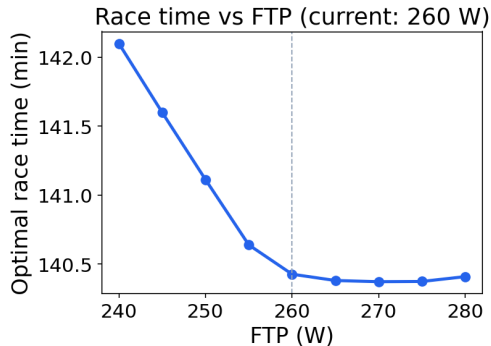
Model 3 — “steady bike, hard run,” derived

Model 3 — optimal pacing: time in zone per leg (total 137' + transitions)



Bike entirely Z3. Energy budget exhausted exactly: 9389 kJ.

Model 3 — what is a watt worth?



+1 CTL = 69–84 s. +1 W $\approx$ 6 s — but only below 260 W.

Model 3 — validated against three real races

Race	CTL	Swim		Bike		Run	
		mdl	act	mdl	act	mdl	act
Spetsathlon '26 (sprint*)	64.9	12.9	13.2	47.3	50.9	18.7	17.7
Epidavros '25 (Olympic)	55.6	27.2	25.8	88.8	91.5	38.9	42.0
70.3 Costa Navarino '25	71.4	33.3	32.9	184.6	180.3	86.4	115.2

minutes · * non-standard 25 km bike

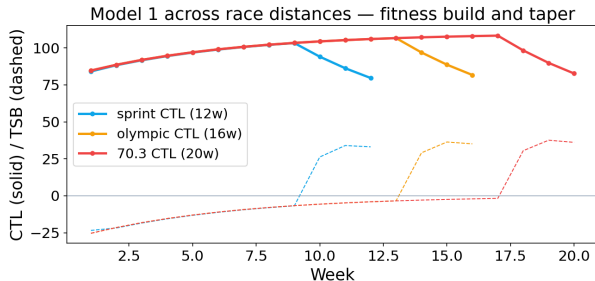
- swim within 0.5 min every time
- binding constraint **switches with distance**: speed caps → energy → infeasible unfed
- the 70.3 run: my **29-minute** wall is the energy constraint, violated in real life

What the bike residual is made of

Bike leg	m/km	no gradient	$\phi(\gamma)$	2-param fit	actual
Spetsathlon (24.4 km)	15.2	43.3	47.3	55.0	50.9
Epidavros (38.7 km)	15.5	78.9	88.8	87.7	91.5
Costa Navarino (88.5 km)	9.1	177.5	184.6	180.0	180.3
total error		23.0 min	10.6 min	8.2 min	
parameters fitted here		0	0	2	

- no gradient $\Rightarrow$ too fast on **every** course: a missing term, not noise
- $\phi(\gamma)$ from physics halves the error with **zero** fitted parameters

Sensitivity across sizes and parameters



- **Sizes:** 12/16/20 weeks, 144–240 vars — structure invariant
- $k_2 \in [1, 3]$: race-day TSB stays 35 ± 2
- **Cost ranging:** plan invariant for $k_1 \in [0.68, 1.56]$, $k_2 \in [1.29, 2.95]$
- **Solve times:** all < 10 ms

Integrating the pipeline: coherence & fixed point

Do the three objectives agree?

- $E = k_E \text{CTL}$ ignores fatigue $\Rightarrow$ four different plans, **same 140.2 min**
- $E = E_0 + k_E(\text{CTL}_T - \lambda \text{ATL}_T)$ is affine in p_T **iff** $\lambda = k_2/k_1$
- $E > 0$ needs $\lambda < 1.75$, so $k_2 = 2$ is *incoherent*; cost ranging licenses $\lambda = 1.5$ — **changing no variable**

Which model comes first? Iterate.

Iter.	Macrocycle	Weeks	Selected A-race	Outcome
1	olympic	16	R14 Costa Navarino (70.3)	re-plan as 70.3
2	70.3	20	R14 Costa Navarino (70.3)	fixed point

What the models actually recommend

- **Train more, much easier.** 6.6 h/wk at 37% hard $\rightarrow$ 9.0 h/wk at **15%**
- **More hours are worthless** above 10.5 h/wk — recovery is what binds
- **Build endurance, not watts.** +1 CTL = 69–84 s; +1 W $\approx$ 0 above 260 W
- **Ride Z3, run hard, fuel 75 g/h.** The 70.3 positive split cost 29 min
- **Protect the calendar, not the budget.** Conflict dual 56.4 vs. 0.094/EUR

- Three coupled models on **real personal data**
- Spanning the syllabus: LP & Simplex, duality & sensitivity, IP & Branch & Bound, network flows & total unimodularity
- The optimizer **re-derives coaching wisdom** — periodization, polarization, “steady bike, hard run”
- Duals name the real bottlenecks: *recovery*, *calendar space*, *endurance*
- Reproducible: `python -m src.run_all`, 60 tests

<https://github.com/manos1503/triathlon-optimization>

Thank you — questions?

Backup: limitations

- **Weekly** time step for the Banister dynamics
- **Sequential** pipeline, not joint optimization
- **Constructed parameters** in Models 2–3 (race values, zone speeds, k_E)
- The LP's +35 TSB over-taper — traced to race-week minimums

Backup: alternative optima & heuristic benchmark

Alternative optima: fix $p_T = p^*$, re-solve min/max total hours

same performance with **124.2 h** or **162.3 h** of season training

vs. a standard coaching heuristic:

- violates the fatigue ceiling in **13 of 16 weeks**
- $p_T = -69.5$ vs. optimal -11.4
- but its race-day TSB $+21$ is inside the coaching band; the LP's $+35$ is not